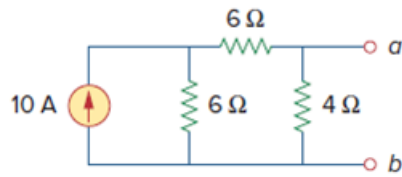


Problem

Find the Thevenin equivalent of the circuit in Fig. 4.112 as seen by looking into terminals a and b .

Figure. 4.112:



Step-by-step solution

Step 1 of 5

Refer to circuit diagram in Figure 4.112 in the text book.

Consider the circuit diagram shown in Figure 1 to calculate the value of the Thevenin's Voltage.

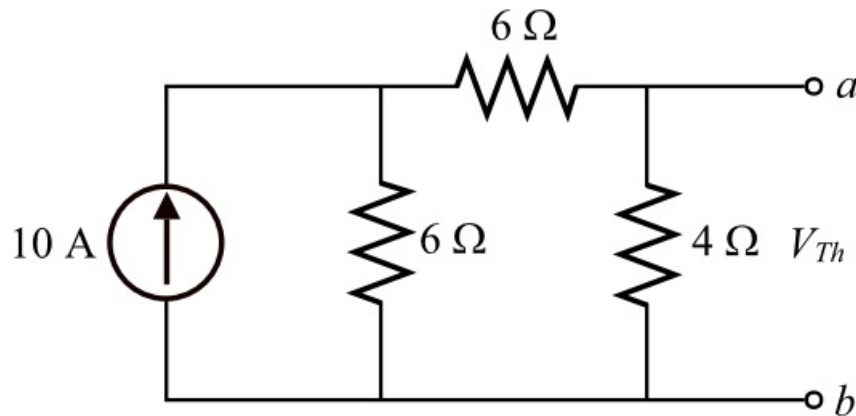


Figure 1

Step 2 of 5

Calculate the current through $4\ \Omega$ resistor using current division rule.

$$\begin{aligned} I_4 &= \left(\frac{6\ \Omega}{6\ \Omega + 6\ \Omega + 4\ \Omega} \right) (10\ \text{A}) \\ &= \left(\frac{6\ \Omega}{16\ \Omega} \right) (10\ \text{A}) \\ &= (0.375)(10\ \text{A}) \\ &= 3.75\ \text{A} \end{aligned}$$

Now, calculate the value of the Thevenin's Voltage V_{Th} .

$$\begin{aligned} V_{Th} &= (4\ \Omega) I_4 \\ &= (4\ \Omega)(3.75\ \text{A}) \\ &= 15\ \text{V} \end{aligned}$$

Therefore, the value of the Thevenin's voltage V_{Th} is, $\boxed{15\ \text{V}}$.

Step 3 of 5

Deactivate the sources from the circuit diagram to calculate the Thevenin's resistance.

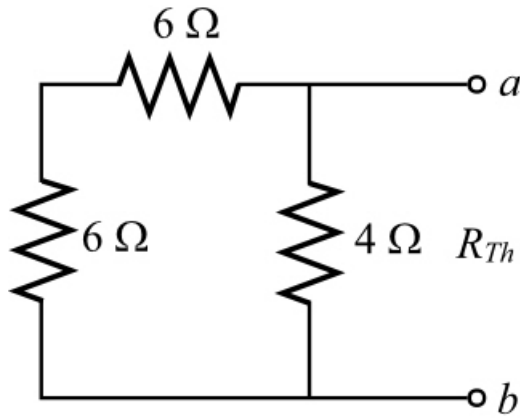


Figure 2

Step 4 of 5

Calculate the value of the Thevenin's resistance R_{Th} .

$$\begin{aligned} R_{Th} &= (4\ \Omega) \parallel (6\ \Omega + 6\ \Omega) \\ &= (4\ \Omega) \parallel (12\ \Omega) \\ &= \frac{(4\ \Omega)(12\ \Omega)}{4\ \Omega + 12\ \Omega} \\ &= \frac{48\ \Omega}{16} \\ &= 3\ \Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance R_{Th} is, $\boxed{3\ \Omega}$.

Step 5 of 5

Draw the Thevenin's equivalent circuit diagram.

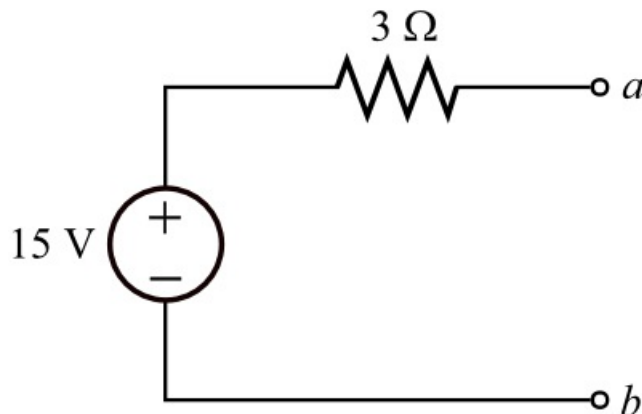


Figure 3: Thevenin's equivalent circuit

Therefore, the Thevenin's equivalent circuit is shown in Figure 3.